

Summary

A transformational leader with a diverse background across collegiate educational institutions, consulting industry, direct sales, and the image consulting industry. Mastered various groups of people from all ethnic backgrounds, sexual orientations, ages, and thinking styles during tenure in education, consulting, and sales. Impacted students by getting to know their needs and helping them outside the classroom when necessary. Excelled in providing customer service to potential and existing customers while working in the image consulting industry and direct sales. Excelled in successfully instructing traditional and nontraditional students, implementing curriculums that cater to all learning styles, developing strong teams, creating growth strategies thru project implementation, sales growth, operations stability, motivational speaking, and mentoring while balancing priorities with each unique client & deliverable.

Education

Ph.D. in Physics—in progress, Alabama A&M University, Normal, AL, expected graduation 05/2025

Master of Science in Physics, Alabama A&M University, Normal, AL

Thesis: Spectral Hole burning Studies and Optical Investigations of Rare-Earth Ion and Dye-Doped Materials

Bachelor of Science in Chemistry, Alabama A&M University, Normal, AL

Honors & Affiliations

- Graduate-Advisor AAMU Society of Physics Students Fall 2024 & Spring 2025
- President, AAMU Society of Physics Students 2024, 2023 and 2022
- ESPSCoR Graduate Research Scholars Program Recipient 2024, 2023, 2022
- Graduate Student At-Large Officer of Alabama Academy of Science 2024
- National Society of Black Physicists Member
- Scholarly Partnership in Nuclear Security (SPINS) Fellow
- National Defense Education Program (NDEP) Scholar
- Sigma Pi Sigma
- American Association Physics Teachers Member
- Member AVS: Science & Technology of Materials, Interfaces and Processing
- Member Delta Sigma Theta Sorority, Inc.

Awards & Recognition

- Alabama A&M University STEM Day 2025 Physics Graduate Winner 1st Place Oral Presentation
- Alabama Academy of Science (AAS) 2025 Poster Competition Graduate, 1st Place Winner and Paper Competition Graduate, 3rd Place Winner
- Alabama Academy of Science 2023 and 2024 Poster Competition Graduate, 2nd Place Winner
- Alabama A&M University STEM Day 2023 Physics Graduate Winner 2nd Place and 2024 Physics Graduate Winner 1st Place
- 2022 BAE Think Tank Awards 1st place Winner \$5000 prize—Novel IR Window Technologies and 2nd place Winner \$3500 prize—Military Vehicle Electrification Advancements in Technology
- 2023 BAE Think Tank Awards 3rd place Winner \$600 prize—Embedded Cyber Resilience
- Recognized at the 2023 Alabama A&M University International Women's History Month Program as

Best Graduate student

- 2023 Moving Science with STEM Stars by Southern Research, 1st Place Winner Graduate Student \$1000 prize

Publications

- II, C., Davis, A., Wilson, L. and Guggilla, P. (2024) Revolutionizing Material Boundaries: High-Performance Diamond-Coated Steel via Long-Term MPCVD. Journal of Minerals and Materials Characterization and Engineering, 12, 334-345. doi: 10.4236/jmmce.2024.126021.
- Clyde Varner, Angela Davis, Ashok K. Batra, Padmaja Guggilla, "Optical band gap engineering and comparison of conductivity of CaTiO₃ and LiNbO₃ doped PVDF films", Results in Optics, Volume 16, 2024, 100720, ISSN 2666-9501, <https://doi.org/10.1016/j.rio.2024.100720>
- Ashok Batra, James Sampson, Angela Davis, James Currie, and Ashok Vaseashta, "Electrospun nanofibers doped with PVDF and PLZT nanoparticles for potential biomedical and energy harvesting applications" J Mater Sci: Mater Electron (2023) 34:1654, <https://www.researchgate.net/publication/373044037>
- <https://ui.adsabs.harvard.edu/abs/2023nsbp.confE.151D/abstract>
- "Strain Engineering of Nano-crystalline CaTiO₃ and LiNbO₃ in PVDF Matrix for efficient smart material" Angela Davis, Clyde Varner, Padmaja Guggilla, Ashok K Batra, (2023), Journal National Society of Black Physicists Conference, Pages 151
- Clyde Varner II, Angela Davis, Leslie Wilson, Padmaja Guggilla, Revolutionizing Material Boundaries: High-Performance Diamond-Coated Steel via Long-Term MPCVD;, Journal of Minerals and Materials Characterization and Engineering, Volume 12, Issue 6, Pages 334-345 (2024)
- In Press: Smart Nanocomposites for Energy Harvesting Biomedical Sensing Applications, <https://link.springer.com/book/9783031848087>
- In Press: Fano Resonance in Ion-Bombarded Au-SiO₂ Nanocomposites: Analysis of Mode Coupling and Optical Properties, Coatings

Internship Experience

Lawrence Livermore National Laboratory (LLNL) | Intern

Summer 2024

- A diamond optically gate field effect transistor (DOG-FET) is being built to meet future grid requirements for higher voltage and current operating devices with higher speeds for improved control, resiliency and efficiency.
- Tasked with the crucial component of identifying and realizing a non-standard etching process.
- Critical etching process required to realize the DOG-FET device.
- Identified etching process and worked towards eliminating issues that arose with this type of fabrication due to the devices micro-scale size.
- Proficiently utilized Scanning Electron Microscope (SEM), Nomarski microscopy, Raman spectroscopy and Photoluminescence (PL)

International Space Weather Camp (ISWC) | Intern

Summer 2023

- South African National Space Agency (SANSA): Dual country internship encompassed travel to South Africa and the University of Alabama Huntsville.
- Utilized Python in South Africa for Geo caching project and analyzed data at UAH.

Scholarly Partnership in Nuclear Security Consortium (SPINS) | Intern

Summer 2021 and 2022

- Traveled to Brookhaven National Laboratory to work on CZT Projects—Development and Testing for

Enhanced Sensitivity and Efficiency to High Energy Gamma Radiation.

- Continued education seminars on nuclear proliferation, nuclear storage and safety standards, radiation detection, and a host of other nuclear and radiation topics.

Technical Skills

- Python
- Raman Spectroscopy
- Scanning Electron Microscope (SEM)
- Nomarski Microscopy
- Fourier transform infrared spectroscopy (FTIR)
- Ultraviolet–visible spectroscopy (UV-VIS)
- Cold Jet Plasma Treatment
- Photoluminescence (PL) spectroscopy

Professional Experience

AAMU Physics Department | Adjunct Instructor Normal, AL August 2024 – Present

Taught algebra based physical science to 90+ students in weekly face to face lecture sessions.

Taught algebra based physical science labs to 90+ students in weekly face to face lab sessions.

J.F. Drake State Community & Technical College Huntsville, AL August 2022-Present Adjunct Physics Professor

Responsible for teaching Physical Science lectures (PHYS1110). This algebra and trigonometry-based course present the essentials of physics, chemistry, earth science, and astronomy. This course shows students how science works, how scientists approach problems, and why science constantly evolves in its search for understanding.

- Teaches weekly for 1.5 hours
- Teaches material from the approved curriculum.
- Develop daily lesson plans to include instructional aids.
- Provided regular and timely feedback to students
- Advise students in matters related to academics, attendance, and behaviors.
- Motivate students to participate in all aspects of the educational process actively.
- Maintained and reported student grades and attendance by college policies.
- Complete professional development and in-service activities by college standards.
- Participate in department curriculum meetings.

Atlanta Technical College Atlanta, GA July 2019-Present Adjunct Physics Professor

Responsible for teaching face to face and online the conceptual Physics lecture (PHYS1110) and the conceptual Physics laboratory (PHYS1110L) courses. These algebra and trigonometry-based courses introduce students to some of the fundamental laws of physics. Topics include systems of units, conversion of units, vector algebra, Newtonian mechanics, fluids and thermodynamics, heat, light, optics, mechanical waves, electricity and magnetism, and modern physics. The laboratory allows the concepts to be conducted in the laboratory setting.

- Teaches weekly for 5 hours
- Ensured proper instrumentation was ordered and available for the department.
- Teaches material from approved curriculum and develops daily lesson plans to include instructional aids.
- Develop daily lesson plans to include instructional aids.

- Assist students in achieving completion of laboratories during class. Provided regular and timely feedback to students
- Advise students in matters related to academics, attendance, and behaviors.
- Motivate students to participate in all aspects of the educational process actively.
- Maintained and report student grades and attendance by university policies.
- Complete professional development and in-service activities by college standards.
- Participate in department curriculum meetings.

Chattahoochee Technical College Atlanta, GA January 2019 - October 2023 Adjunct Physics Professor

Responsible for teaching the conceptual Physics lecture (PHYS1110) and the conceptual Physics laboratory (PHYS1110L) courses. These algebra and trigonometry-based courses introduce students to some of the fundamental laws of physics. Topics include systems of units, conversion of units, vector algebra, reaction time, oscillations, pendulums, refraction and reflection, heat and optics, and Hooke's Law. The laboratory allows the concepts to be conducted in the laboratory setting allowing students to understand the concepts better.

- Teaches weekly for 1.5 hours
- Teaches material from approved curriculum and develops daily lesson plans to include instructional aids.
- Ensure necessary equipment is present for each laboratory station.
- Develop daily lesson plans to include instructional aids.
- Assist students in achieving completion of laboratories during class.
- Provide regular and timely feedback to students.
- Advise students in matters related to academics, attendance, and behaviors.
- Motivate students to participate in all aspects of the educational process actively.
- Maintained and report student grades and attendance by university policies.
- Complete professional development and in-service activities by college standards.
- Participate in department curriculum meetings.

Georgia State University Perimeter College Decatur, GA, January 2016-July, 2018 Physical Science Laboratory Supervisor

Executed the capacity of Physical Science Laboratory Supervisor. I created and spearheaded the implementation of a new methodology for all laboratory equipment locations and storage. The new method increased instructors' instruction time and assisted students by boosting their equipment knowledge, confidence in a laboratory environment, and equipment setup during class. I also ensured that the required laboratory supplies and equipment were maintained, ordered, and prepared weekly for the Astronomy, Geology, and Physics labs.

- Maintained accurate records and participated in laboratory course curriculum development.
- Maintained, revised, and provided procedures manual regarding lab policies and supplementary lab materials for full and part-time laboratory instructors.
- Maintained a comprehensive collection of up-to-date material safety data sheets (MSDS) for all chemicals in the area of responsibility.
- Maintained a comprehensive Astronomy, Geology, and Physics inventory.
- Established procedures for instrumentation/equipment usage.
- Prepared schedules for Astronomy, Geology, and Physics lab sections each semester.
- Instructed 2-3 courses (6-9 course workload hours) per semester.
- Taught material from approved curriculum and developed daily lesson plans to include instructional aids.
- Developed daily lesson plans to include instructional aids.
- Assisted students in achieving completion of laboratories during class. Provided regular and timely feedback to students.

- Advised students in matters related to academics, attendance, and behaviors.
- Motivated students to actively participate in all aspects of the educational process.
- Maintained and reported student grades and attendance by university policies.
- Completed professional development and in-service activities by college standards.
- Participated in department curriculum meetings.
- Participated in department-sponsored events during the year. These events are subject to but not limited to: Department sponsored guest lecture series and Atlanta Science Week Activities.

Georgia Perimeter College Decatur, GA, March 2009-December 2015 Adjunct Physics Professor

Instructed the Physics 1111 laboratory and the Physics 2211 laboratory. These courses were algebra and trigonometry-based laboratory courses that covered the following areas, which were subject to but not limited to Measurements, Newton's Second Law, Buoyancy, Conservation of Energy, and Uniform Circular Motion.

- Created a syllabus that covered the curriculum and all expectations for the class and students during the quarter.
- Managed students, laboratories, and equipment during class.
- Assisted students in achieving the completion of laboratory objectives.
- Provided regular and timely feedback to students.
- Advised students in matters related to academics, attendance, and behaviors.
- Motivated students to actively participate in all aspects of the educational process.
- Maintained and reports student grades and attendance by company policies.
- Participated in required department meetings.

ITT Technical College Atlanta, GA, December 2011-March 2015 Adjunct Professor of Science

Instructed the general science courses encompassing biology, chemistry, physics, and physical science. The basic concepts of all disciplines listed above were taught. I was also responsible for instructing students from a non-traditional college background and ensuring their ability to learn the sciences, make presentations based on the coursework, and improve their public speaking skills.

- Taught material from approved curriculum and developed daily lesson plans to include instructional aids.
- Developed daily lesson plans to include instructional aids.
- Assisted students in achieving the completion of objectives.
- Provides regular and timely feedback to students.
- Participated in school retention initiatives by maintaining productive contact with students and getting in touch with and assisting absent students.
- Advised students in matters related to academics, attendance, and behaviors.
- Motivated students to actively participate in all aspects of the educational process.
- Maintained and reports student grades and attendance by company policies.
- Completed professional development and in-service activities by college standards.
- Participated in department meetings.

Rainbow Park GED Program, Decatur, GA, March 2009-December 2014 Science & Mathematics Instructor

As the head of the math and science GED Program, I created class curriculum, lesson plans, and classroom environments that catered to all student learning styles and ensured students learned and completed assignments. Focusing on learning styles and implementing additional study sessions based on how students learned increased their ability to pass the GED exam; this also ensured a nurturing environment that allowed students to overcome insecurities around math and science.

- Created curriculum for Basic Mathematics, Algebra I, Algebra II, and Science Classes.
- Taught students Algebra I & II.
- Taught Science and Social Studies.
- Facilitated student learning through in-class instruction and one on one tutoring.
- Conducted research in the required areas of mathematics and science.
- Assigned and checked assignments.
- Tested and ensured students' growth and knowledge of subjects.
- Developed and implemented programs to encourage student participation.
- Supervised others in the maintenance and enforcement of program policies.
- Assisted in the coordination and implementation of program schedule and curriculum.

Mary Kay Cosmetics, Atlanta, GA—Director, December 2004-2016

As a Mary Kay Sales Director, I was a public speaker, motivator, and recruiter that trained and mentored my recruits and over 200 women in our national area. I was responsible for creating and distributing a monthly schedule that ensured all directors had an opportunity to teach and that all classes were correctly rotated. The pieces of training I designed and taught are Business communication, financial management, Internet marketing, Recruiting, Merchandising, Sales techniques and psychology, Money Management, Internet Marketing, Organization, Image and Time management, personality classes, fiscal responsibilities, product knowledge, online marketing, Incorporating social media in your business and Sales 101. I displayed the ability to work independently under pressure toward daily, weekly, and monthly deadlines.

- Develops and delivers classroom-style training for the National area related to building and maintaining a home-based business for independent Direct Sales Professionals
- Responsible for planning, directing, and organizing events and activities.
- These responsibilities include creating monthly schedules for 20-plus directors and determining which classes.
- Plans and conducts group and individual coaching sessions with regional sales force members to build and develop home-based businesses for immediate income or long-term, career-oriented business development.
- Delivers marketing presentations to large groups to recruit potential sales and business development consultants.
- A self-motivated business coach that helps team members improve and maintain both interpersonal and business skills.
- Recruiting and maintaining a sales base.
- Recruiting and training prospective consultants based on their individual goals
- Motivational Speaker

Accenture

Six years of experience within the IT Consulting and management consulting industry. My focus was both requirements gathering and quality assurance testing. Leveraged problem-solving skills and analytical skills acquired from Chemistry and Physics background to develop IT solutions through requirements gathering, business case validation, and developing accurate deliverables. These problem-solving skills

were utilized in producing requirements deliverables, enabling testing and development to deliver new products and services, and led team members through integrated testing processes while implementing new products, product launches, and services. Requirements gathering and quality assurance testing enabled the following skills to develop. Excelled at detailed oriented and proficient at data collection and analysis, requirements planning/research, and project planning. My expertise was also valuable during testing, quality assurance, product rollout, and post-product rollout support.

Atlanta, GA – Functional Analyst, August 2002 – September 2006

- Responsible for managing a team and distributing the workload as necessary to ensure all

deadlines were met.

- Provided business consulting services through customer identification, evaluation, implementation, and recommended solutions that solve the customer's strategic business goals.
- Analyzed organizations and designed processes and systems, assessing the business model and its integration with technology.
- Assessed the current state, identified customer requirements, and defined the future state and business solution.
- Spearheaded information-gathering sessions, analyzing, documenting, and editing requirements, specifications, and business processes from business leaders and IT stakeholders.
- Elicited, analyzed, and validated project requirements that resulted in new applications or enhancements to existing applications, systems, and business process design.
- Spearheaded requirement-gathering efforts to create universal templates used throughout the project, which assisted in reducing writing time by 10%.
- Created an environment of productivity and personal accountability that was instrumental in reducing requirement writing time.
- Designated and trained a five-person staff on using requirement-gathering templates and capturing more accurate revisions during requirement-gathering sessions.
- Coordinated development effort of deliverables across multiple teams to ensure accurate research and learning topics, timely delivery of deliverables, and client feedback was incorporated and created high-quality deliverables.
- Coordinated development effort of deliverables across teams to ensure researching and learning topics, timely delivery of deliverables, incorporating client feedback, creating high-quality deliverables, and incorporating development/testing feedback.
- Created Application Requirements Specifications documents, allowing development and testing to understand the business needs better.
- Liaison between development, testing, and BellSouth business representatives to resolve and receive clarification around questions and issues.
- Utilized in-depth functional knowledge of BellSouth's business needs as they pertained to the requirements which drove the initiative before its release into production. Reviewed requirements to determine if gaps existed between the company's desires and the application's limitations.
- Organized and led testing efforts in which the primary system foundation was a sales order capture and customer support application using the Oracle Customer Relationship Management (CRM) software.
- Effectively communicated issues encountered during requirement reviews and testing to the business.

Atlanta, GA – Systems Test Analyst, May 2000 – July 2002

Organized and led testing efforts in which the primary system foundation was a sales order capture and customer support application using the Oracle 11i Customer Relationship Management (CRM) software.

- Developed and executed test scenarios according to written test specifications and requirements.
- Determined what impact (if any) defect fixes would have on the environments before Performance Testing.
- Recreated and effectively communicated defects and issues encountered during assembly testing to the development team and business personnel.